

Problem 3.1 – Sample Geometry

Given the data matrix

$$X = \begin{bmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{bmatrix}.$$

Thus, the three observations are

$$(9, 1), \quad (5, 3), \quad (1, 2).$$

The two variable vectors are

$$\mathbf{y}_1 = \begin{bmatrix} 9 \\ 5 \\ 1 \end{bmatrix}, \quad \mathbf{y}_2 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}.$$

(a) Scatter plot in $p = 2$ dimensions

The sample mean is

$$\bar{x}_1 = \frac{9 + 5 + 1}{3} = 5,$$

and

$$\bar{x}_2 = \frac{1 + 3 + 2}{3} = 2.$$

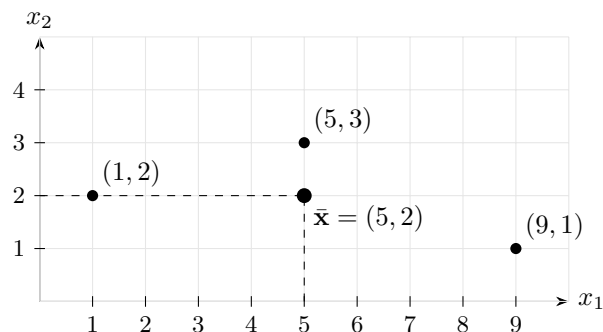
Therefore,

$$\bar{\mathbf{x}} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

and the sample mean is the point

$$(5, 2).$$

The scatter plot is shown below.



(b) $n = 3$ -dimensional representation

The two variables can be represented as vectors in $\mathbb{R}^3$:

$$\mathbf{y}_1 = \begin{bmatrix} 9 \\ 5 \\ 1 \end{bmatrix}, \quad \mathbf{y}_2 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}.$$

The sample means are

$$\bar{x}_1 = 5, \quad \bar{x}_2 = 2.$$

Thus,

$$\bar{x}_1 \mathbf{1} = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}, \quad \bar{x}_2 \mathbf{1} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}.$$

The deviation vectors are

$$\mathbf{y}_1 - \bar{x}_1 \mathbf{1} = \begin{bmatrix} 9 \\ 5 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} = \boxed{\begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}},$$

and

$$\mathbf{y}_2 - \bar{x}_2 \mathbf{1} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \boxed{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}}.$$

Both deviation vectors satisfy

$$z_1 + z_2 + z_3 = 0.$$

Indeed,

$$4 + 0 - 4 = 0$$

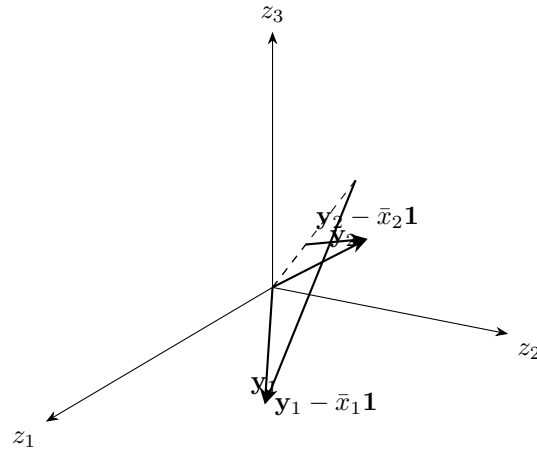
and

$$-1 + 1 + 0 = 0.$$

Therefore, they lie in the plane

$$z_1 + z_2 + z_3 = 0.$$

A 3-dimensional sketch is shown below.



The diagram shows the original variable vectors together with the deviation vectors translated so that they start at their respective mean vectors.

(c) Deviation vectors from the origin

For purposes of calculating their lengths and the angle between them, we translate the deviation vectors so that both emanate from the origin.

The two vectors are

$$\mathbf{d}_1 = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}, \quad \mathbf{d}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

Their lengths are

$$\begin{aligned} \|\mathbf{d}_1\| &= \sqrt{4^2 + 0^2 + (-4)^2} \\ &= \sqrt{32} \\ &= \boxed{4\sqrt{2}}, \end{aligned}$$

and

$$\begin{aligned} \|\mathbf{d}_2\| &= \sqrt{(-1)^2 + 1^2 + 0^2} \\ &= \boxed{\sqrt{2}}. \end{aligned}$$

Their dot product is

$$\mathbf{d}_1' \mathbf{d}_2 = (4)(-1) + (0)(1) + (-4)(0) = -4.$$

Therefore,

$$\cos \theta = \frac{\mathbf{d}_1' \mathbf{d}_2}{\|\mathbf{d}_1\| \|\mathbf{d}_2\|}.$$

Substituting,

$$\cos \theta = \frac{-4}{(4\sqrt{2})(\sqrt{2})} = -\frac{1}{2}.$$

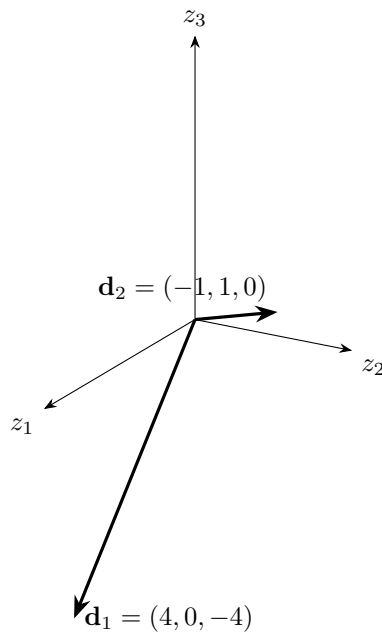
Hence,

$$\cos \theta = -\frac{1}{2}$$

and therefore

$$\theta = 120^\circ.$$

The vectors can be sketched from the origin as follows.



The lengths and angle have direct statistical interpretations.

The sample covariance matrix is

$$S_n = \frac{1}{n-1} D' D.$$

The diagonal elements are related to the squared lengths:

$$S_{11} = \frac{\|\mathbf{d}_1\|^2}{n-1},$$

and

$$S_{22} = \frac{\|\mathbf{d}_2\|^2}{n-1}.$$

Since $n = 3$,

$$S_{11} = \frac{(4\sqrt{2})^2}{2} = \frac{32}{2} = 16,$$

and

$$S_{22} = \frac{(\sqrt{2})^2}{2} = \frac{2}{2} = 1.$$

The off-diagonal element is

$$S_{12} = \frac{\mathbf{d}_1' \mathbf{d}_2}{n-1} = \frac{-4}{2} = -2.$$

Therefore,

$$S_n = \begin{bmatrix} 16 & -2 \\ -2 & 1 \end{bmatrix}.$$

The correlation matrix is

$$R = \begin{bmatrix} 1 & r_{12} \\ r_{12} & 1 \end{bmatrix},$$

where

$$r_{12} = \frac{S_{12}}{\sqrt{S_{11}S_{22}}}.$$

Thus,

$$r_{12} = \frac{-2}{\sqrt{16(1)}} = -\frac{1}{2}.$$

Therefore,

$$R = \begin{bmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix}.$$

Consequently,

$$r_{12} = \cos \theta = -\frac{1}{2}.$$

Hence, the angle between the two centered variable vectors is

$$\theta = 120^\circ.$$

Problem 3.6 – Deviations and Generalized Variance

Given

$$X = \begin{bmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{bmatrix}.$$

Here,

$$n = 3, \quad p = 3.$$

(a) Matrix of deviations and rank

The sample means are

$$\bar{x}_1 = \frac{-1 + 2 + 5}{3} = 2,$$

$$\bar{x}_2 = \frac{3 + 4 + 2}{3} = 3,$$

and

$$\bar{x}_3 = \frac{-2 + 2 + 3}{3} = 1.$$

Thus,

$$\bar{\mathbf{x}} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}.$$

The centered matrix is

$$D = X - \mathbf{1}\bar{\mathbf{x}}'.$$

Therefore,

$$D = \begin{bmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 3 & 1 \\ 2 & 3 & 1 \\ 2 & 3 & 1 \end{bmatrix}.$$

Hence,

$$D = \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix}.$$

Because every centered column sums to zero,

$$\text{rank}(D) \leq n - 1 = 2.$$

Since the first two columns are linearly independent,

$$\text{rank}(D) = 2.$$

Thus, D is not of full rank.

(b) Sample covariance matrix and generalized sample variance

The sample covariance matrix is

$$S = \frac{1}{n-1} D' D.$$

Since $n = 3$,

$$S = \frac{1}{2} D' D.$$

We have

$$D' = \begin{bmatrix} -3 & 0 & 3 \\ 0 & 1 & -1 \\ -3 & 1 & 2 \end{bmatrix}.$$

Thus,

$$D' D = \begin{bmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{bmatrix}.$$

Therefore,

$$S = \begin{bmatrix} 9 & -\frac{3}{2} & \frac{15}{2} \\ -\frac{3}{2} & 1 & -\frac{1}{2} \\ \frac{15}{2} & -\frac{1}{2} & 7 \end{bmatrix}.$$

Since

$$\text{rank}(S) \leq \text{rank}(D) = 2 < 3,$$

S is singular. Hence,

$$|S| = 0.$$

Geometrically, the generalized variance is proportional to the squared volume of the covariance ellipsoid. Since

$$|S| = 0,$$

the ellipsoid has zero three-dimensional volume.

Therefore, the observations lie in a lower-dimensional subspace.

The generalized variance is zero because the data have rank 2.

(c) Total sample variance

The total sample variance is

$$\text{tr}(S) = S_{11} + S_{22} + S_{33}.$$

Therefore,

$$\text{tr}(S) = 9 + 1 + 7 = \boxed{17}.$$

Thus,

$$\boxed{\text{Total sample variance} = 17}.$$

Problem 3.9 – Linearly Dependent Test Scores

The data matrix is

$$X = \begin{bmatrix} 12 & 17 & 29 \\ 18 & 20 & 38 \\ 14 & 16 & 30 \\ 20 & 18 & 38 \\ 16 & 19 & 35 \end{bmatrix}.$$

Here,

$$n = 5, \quad p = 3.$$

(a) Mean-corrected data matrix and linear dependence

The column means are

$$\bar{x}_1 = \frac{12 + 18 + 14 + 20 + 16}{5} = 16,$$

$$\bar{x}_2 = \frac{17 + 20 + 16 + 18 + 19}{5} = 18,$$

and

$$\bar{x}_3 = \frac{29 + 38 + 30 + 38 + 35}{5} = 34.$$

Thus,

$$\bar{\mathbf{x}} = \begin{bmatrix} 16 \\ 18 \\ 34 \end{bmatrix}.$$

The mean-corrected matrix is

$$D = X - \mathbf{1}\bar{\mathbf{x}}'.$$

Hence,

$$D = \begin{bmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix}.$$

Let the columns be $\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3$. We observe that

$$\mathbf{d}_3 = \mathbf{d}_1 + \mathbf{d}_2.$$

Therefore,

$$\mathbf{d}_1 + \mathbf{d}_2 - \mathbf{d}_3 = \mathbf{0}.$$

A vector establishing the linear dependence is

$$a = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

Equivalently,

$$Da = 0.$$

(b) Sample covariance matrix and generalized variance

The sample covariance matrix is

$$S = \frac{1}{n-1} D' D.$$

Since $n = 5$,

$$S = \frac{1}{4} D' D.$$

Computing,

$$D' D = \begin{bmatrix} 40 & 12 & 52 \\ 12 & 10 & 22 \\ 52 & 22 & 74 \end{bmatrix}.$$

Therefore,

$$S = \begin{bmatrix} 10 & 3 & 13 \\ 3 & \frac{5}{2} & \frac{11}{2} \\ 13 & \frac{11}{2} & \frac{37}{2} \end{bmatrix}.$$

Since D does not have full column rank,

$$\text{rank}(S) < 3,$$

and hence

$$\boxed{|S| = 0}.$$

Now,

$$Sa = \begin{bmatrix} 10 & 3 & 13 \\ 3 & \frac{5}{2} & \frac{11}{2} \\ 13 & \frac{11}{2} & \frac{37}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

The components are

$$10 + 3 - 13 = 0,$$

$$3 + \frac{5}{2} - \frac{11}{2} = 0,$$

and

$$13 + \frac{11}{2} - \frac{37}{2} = 0.$$

Therefore,

$$\boxed{Sa = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}.$$

Thus, a is an eigenvector corresponding to

$$\boxed{\lambda = 0}.$$

(c) Verify the third column is the sum of the first two

The first two columns are

$$\mathbf{x}_1 = \begin{bmatrix} 12 \\ 18 \\ 14 \\ 20 \\ 16 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 17 \\ 20 \\ 16 \\ 18 \\ 19 \end{bmatrix}.$$

Their sum is

$$\mathbf{x}_1 + \mathbf{x}_2 = \begin{bmatrix} 29 \\ 38 \\ 30 \\ 38 \\ 35 \end{bmatrix}.$$

But this is exactly

$$\mathbf{x}_3 = \begin{bmatrix} 29 \\ 38 \\ 30 \\ 38 \\ 35 \end{bmatrix}.$$

Therefore,

$$\boxed{\mathbf{x}_3 = \mathbf{x}_1 + \mathbf{x}_2}.$$

Hence,

$$\boxed{a_1 = 1, \quad a_2 = 1, \quad a_3 = -1}.$$

Problem 3.18

Given Data: Mean vector $\bar{x} = [0.766 \quad 0.508 \quad 0.438 \quad 0.161]'$ and covariance matrix S .

(a) Total Energy Consumption

Define the transformation vector $a = \mathbf{1} = [1 \quad 1 \quad 1 \quad 1]'$. Total energy mean:

$$\bar{y} = a' \bar{x} = 0.766 + 0.508 + 0.438 + 0.161 = 1.873 = \frac{1873}{1000}$$

The total variance is the sum of all elements in the 4×4 matrix S :

$$s_y^2 = \mathbf{1}' S \mathbf{1} = \sum_{i=1}^4 \sum_{k=1}^4 s_{ik} = 1.760 + 1.398 + 0.510 + 0.245 = 3.913 = \frac{3913}{1000}$$

Final Answer (a): The sample mean of total energy is 1.873 and the variance is 3.913.

(b) Excess Petroleum Over Natural Gas

Define the transformation vector $c = [1 \quad -1 \quad 0 \quad 0]'$. The mean of the excess ($v = x_1 - x_2$) is:

$$\bar{v} = c' \bar{x} = 0.766 - 0.508 = 0.258 = \frac{129}{500}$$

The sample variance is:

$$s_v^2 = c' S c = s_{11} + s_{22} - 2s_{12} = 0.856 + 0.568 - 2(0.635) = 0.154 = \frac{77}{500}$$

The sample covariance with the total variable y is computed via $c' S \mathbf{1}$, which equals the sum of the first row of S minus the sum of the second row:

$$\text{Cov}(v, y) = \sum_{j=1}^4 s_{1j} - \sum_{j=1}^4 s_{2j} = 1.760 - 1.398 = 0.362 = \frac{181}{500}$$

Final Answer (b): The mean excess is 0.258, the variance is 0.154, and the covariance with total energy is 0.362.

Problem 3.20

Given Data: Dataset of $n = 25$ incidents.

(a) Summary Statistics Method

Calculating sums from the table: $\sum x_1 = 281.9$ and $\sum x_2 = 435.4$. The individual variable means are:

$$\bar{x}_1 = \frac{281.9}{25} = 11.276, \quad \bar{x}_2 = \frac{435.4}{25} = 17.416$$

The mean of the difference $d = x_2 - x_1$ is $\bar{d} = 17.416 - 11.276 = 6.14$.

To calculate the variance of the difference from summary statistics, we need s_{11} , s_{22} , and s_{12} :

$$\begin{aligned} \sum x_1^2 = 3732.53 &\implies s_{11} = \frac{3732.53 - 25(11.276)^2}{24} = \frac{553.8156}{24} \\ \sum x_2^2 = 9604.20 &\implies s_{22} = \frac{9604.20 - 25(17.416)^2}{24} = \frac{2021.2736}{24} \\ \sum x_1 x_2 = 4861.90 &\implies s_{12} = \frac{4861.90 - 25(11.276)(17.416)}{24} = \frac{-47.668}{24} \end{aligned}$$

Using the linear combination variance formula $s_d^2 = s_{11} + s_{22} - 2s_{12}$:

$$s_d^2 = \frac{553.8156 + 2021.2736 - 2(-47.668)}{24} = \frac{2670.44}{24} = \frac{66761}{600} \approx 111.268$$

Final Answer (a): Mean difference is exactly $\frac{307}{50} = 6.14$, and the variance is exactly $\frac{66761}{600} \approx 111.268$.

(b) Individual Values Method

We transform the bivariate data into a univariate dataset $d_j = x_{j2} - x_{j1}$ for all $j = 1, \dots, 25$. Summing the individual differences gives $\sum d_j = 153.5$.

$$\text{Mean } (\bar{d}) = \frac{153.5}{25} = 6.14$$

Summing the squared differences gives $\sum d_j^2 = 3612.93$.

$$\text{Variance } (s_d^2) = \frac{\sum d_j^2 - 25(\bar{d})^2}{24} = \frac{3612.93 - 25(6.14)^2}{24} = \frac{2670.44}{24} = \frac{66761}{600}$$

Final Answer (b): Both the mean and variance computed directly from the individual values match the summary statistics methodology exactly. Mean = 6.14 and Variance = $\frac{66761}{600}$.